

Removing the full-preimage sigma-algebra hypothesis: an exact transfer-operator characterization of k -quasi n -power posinormal weighted composition operators

Candidate full solution packet for arXiv:2507.06511

12 August 2026

Abstract

The first future-work direction in arXiv:2507.06511 asks for the assumption $\varphi^{-j}(\Sigma) = \Sigma$ to be relaxed in the composition and weighted composition results. We remove it completely. For an arbitrary nonsingular transformation inducing a bounded composition operator, the Perron transfer operators $\mathcal{P}_j = (C_\varphi^*)^j$ give exact necessary-and-sufficient conditions. For the weighted composition operator itself, the conditions are scalar almost-everywhere inequalities in both regimes $k \geq n$ and $k < n$; the latter follows from a sharp weighted transfer lemma. For the adjoint, the condition is an exact one-step compressed transfer inequality. The results allow noninvertible transformations and complex weights and specialize to the unweighted case by setting the weight equal to one.

1 Source question and precise scope

The source is D. Dhara, P. Pattanayak, R. Rattan, and S. Sarkar, “ k -quasi n -power posinormal Weighted Composition and Cauchy Dual of Moore–Penrose inverse of Lambert Operators,” arXiv:2507.06511, Section 6, physical PDF page 15. Its first future direction reads exactly:

“Several directions emerge naturally from this work. First, relaxing the assumption $T^{-k}(\Sigma) = \Sigma$ would extend the applicability of the results to non-measure-preserving transformations.”

The rendered source-page evidence is reproduced below.

Therefore, taking $\lambda \geq 2^n$, S_μ is k -quasi n -power posinormal for $k > n + 1$.

6. CONCLUSION AND FUTURE WORK

This paper has systematically investigated the class of k -quasi n -power posinormal operators across diverse settings. For composition operators C_T on $L^2(\mu)$, we established necessary and sufficient conditions for k -quasi n -power posinormality, expressed via Radon-Nikodym derivatives h_k and the transformation T . Analogous characterizations were derived for weighted composition operators W_T , involving products of weights π_k and shifts of h_k . For Lambert conditional operators $T = M_w E M_u$, we characterized when their Cauchy duals $\omega(T)$ —constructed via the Moore-Penrose inverse—are k -quasi n -power posinormal, yielding the unified condition $\lambda^2 E(u^2)^{2n-3} E(w^2)^{2n-3} \geq E(uw)^{2n-4}$ on K . Finally, we extended these results to weighted shifts S_μ on rooted directed trees and provided explicit examples validating the theory. These contributions generalize classical posinormal operator theory and bridge measure theory, conditional expectations and operator theory.

Several directions emerge naturally from this work. First, relaxing the assumption $T^{-k}(\Sigma) = \Sigma$ would extend the applicability of the results to non-measure-preserving transformations. Second, exploring the class beyond L^2 -spaces—such as in L^p ($p \neq 2$), non-commutative L^2 -spaces of von Neumann algebras, or Hardy/Bergman spaces—could reveal deeper connections to harmonic analysis. Third, investigating applications in frame theory (e.g., optimal frame bounds via Cauchy duals), quantum information (e.g., posinormal quantum channels), or network dynamics (e.g., stability of flows modelled by weighted shifts on trees) would demonstrate the practical utility of these operators. Additionally, extending the framework to broader operator classes (e.g., Toeplitz operators, matrix-valued weighted shifts) or linking it to Aluthge transformations and subnormality remains open. Computational methods to verify posinormality for high-dimensional operators could also be developed.

The sigma-algebra condition and measure preservation are logically distinct: the former is a full-preimage/bimeasurability condition. The theorem below removes precisely that sigma-algebra hypothesis while retaining the source's standard nonsingularity and boundedness framework. It covers both ordinary and weighted composition operators, as well as their adjoints, which are the subjects of Sections 2–3 of the source.

2 Notation and transfer facts

Let (X, Σ, μ) be a complete sigma-finite measure space, let $\varphi : X \rightarrow X$ be nonsingular, and suppose

$$C = C_\varphi : L^2(\mu) \rightarrow L^2(\mu), \quad Cf = f \circ \varphi,$$

is bounded. Let $\pi \in L^\infty(\mu)$ and $W = M_\pi C$. No assumption $\varphi^{-j}(\Sigma) = \Sigma$ is made.

For $j \geq 0$, put

$$\pi_0 = 1, \quad \pi_j = \prod_{i=0}^{j-1} \pi \circ \varphi^i, \quad \mathcal{P}_j = (C^*)^j.$$

On L^2 , $\mathcal{P}_j$ is the adjoint transfer operator. On nonnegative measurable functions it has its usual monotone extension, characterized by

$$\int_X \mathcal{P}_j(a) \bar{g} d\mu = \int_X a \overline{g \circ \varphi^j} d\mu \quad (2.1)$$

whenever the two sides are defined. In particular,

$$\mathcal{P}_j(a(g \circ \varphi^j)) = g\mathcal{P}_j(a), \quad \int_X \mathcal{P}_j(a) d\mu = \int_X a d\mu. \quad (2.2)$$

Define the nonnegative functions

$$q_j := \mathcal{P}_j(|\pi_j|^2), \quad q_0 = 1. \quad (2.3)$$

Thus q_j is the Radon–Nikodym density of $A \mapsto \int_{\varphi^{-j}(A)} |\pi_j|^2 d\mu$.

We use the extended quotient convention

$$\frac{|b|^2}{q} = 0 \quad \text{on } \{b = 0 = q\}, \quad \frac{|b|^2}{q} = +\infty \quad \text{on } \{q = 0, b \neq 0\}. \quad (2.4)$$

The following sharp lemma is the mechanism that turns the noninvertible $k < n$ case into a scalar condition.

Lemma 1 (sharp weighted transfer bound). *Let $r \geq 0$, let $q \geq 0$, and let b be measurable, with the displayed expressions well-defined by truncation. For $\lambda \geq 0$, the following are equivalent:*

$$\|\mathcal{P}_r(bf)\|_2^2 \leq \lambda^2 \int_X q|f|^2 d\mu \quad \text{for every admissible } f, \quad (2.5)$$

$$\mathcal{P}_r\left(\frac{|b|^2}{q}\right) \leq \lambda^2 \quad \text{almost everywhere.} \quad (2.6)$$

Consequently, the optimal squared constant in (2.5) is $\|\mathcal{P}_r(|b|^2/q)\|_\infty$, possibly $+\infty$.

Proof. The transfer Cauchy–Schwarz inequality and (2.2) give

$$|\mathcal{P}_r(bf)|^2 \leq \mathcal{P}_r\left(\frac{|b|^2}{q}\right) \mathcal{P}_r(q|f|^2).$$

Thus (2.6), integration, and (2.2) imply (2.5).

Conversely, write $a = |b|^2/q$. For bounded g of finite-measure support, first on truncation sets where all quantities are bounded, take

$$f = \frac{\bar{b}}{q}(g \circ \varphi^r).$$

Then (2.2) yields

$$\mathcal{P}_r(bf) = g\mathcal{P}_r(a), \quad \int_X q|f|^2 d\mu = \int_X |g|^2 \mathcal{P}_r(a) d\mu.$$

Substitution in (2.5) gives

$$\int_X |g|^2 \mathcal{P}_r(a)^2 d\mu \leq \lambda^2 \int_X |g|^2 \mathcal{P}_r(a) d\mu.$$

Arbitrary such g force $\mathcal{P}_r(a) \leq \lambda^2$ almost everywhere. Monotone truncation proves the assertion with the convention (2.4). $\square$

3 Exact characterization without the sigma-algebra hypothesis

Recall that an operator A is k -quasi n -power posinormal if some $\lambda > 0$ satisfies

$$A^{*k}(\lambda^2 A^* A - A^n A^{*n})A^k \geq 0. \quad (3.1)$$

Theorem 2 (full transfer characterization). *Let $k \geq 0$ and $n \geq 1$.*

(A) The weighted composition operator. *The operator $W = M_\pi C_\varphi$ is k -quasi n -power posinormal with constant λ if and only if the following corresponding condition holds.*

If $k \geq n$, with $r = k - n$, then

$$\mathcal{P}_r(q_n^2 |\pi_r|^2) \leq \lambda^2 q_{k+1} \quad a.e. \quad (3.2)$$

If $k < n$, with $r = n - k$, then

$$\mathcal{P}_r\left(\frac{|\pi_r|^2 q_k^2}{q_{k+1}}\right) \leq \lambda^2 \quad a.e. \quad (3.3)$$

with the convention (2.4). Both conditions give the optimal squared constant by taking the essential supremum of the left/right ratio in (3.2), or of the left side in (3.3).

(B) The adjoint. *Let $S_k = \text{supp } q_k$ modulo null sets. The adjoint W^* is k -quasi n -power posinormal with constant λ if and only if*

$$\int_X q_n |g|^2 d\mu \leq \lambda^2 \|\mathcal{P}_1(\bar{\pi}g)\|_2^2 \quad \text{for every } g \in L^2(S_k). \quad (3.4)$$

Equivalently, on $L^2(\mu)$,

$$M_{1_{S_k} q_n} \leq \lambda^2 M_{1_{S_k}}, WW^*, M_{1_{S_k}}, \quad WW^* = M_\pi C_\varphi \mathcal{P}_1 M_{\bar{\pi}}. \quad (3.5)$$

Thus (3.4) is an exact one-step transfer criterion involving only q_n , the support of q_k , and the original weighted composition operator.

Proof intuition

All higher powers have a simple transfer normal form: $W^j = M_{\pi_j} C^j$ and $W^{*j} W^j = M_{q_j}$. In the regime $k \geq n$, the middle $W^{*n} W^n$ collapses to a multiplier, leaving another multiplier after one transfer. In the regime $k < n$, the uncanceled adjoint power leaves the map $\mathcal{P}_{n-k} M_b$; Lemma 1 computes its exact norm relative to the denominator q_{k+1} . For W^* , the only vectors that matter are in $\overline{\text{Ran } W^{*k}} = L^2(S_k)$, reducing every power condition to a one-step lower transfer bound on that support.

Proof of Theorem 2. From the definitions and (2.2),

$$W^j = M_{\pi_j} C^j, \quad W^{*j} W^j = C^{*j} M_{|\pi_j|^2} C^j = M_{q_j}. \quad (3.6)$$

Condition (3.1) for W is equivalent to

$$\|W^{*n} W^k f\|_2^2 \leq \lambda^2 \|W^{k+1} f\|_2^2 = \lambda^2 \int_X q_{k+1} |f|^2 d\mu. \quad (3.7)$$

Suppose first that $k \geq n$ and set $r = k - n$. Using (3.6),

$$W^{*n}W^k = W^{*n}W^nW^r = M_{q_n\pi_r}C^r.$$

Hence

$$(W^{*n}W^k)^*(W^{*n}W^k) = C^{*r}M_{q_n^2|\pi_r|^2}C^r = M_{\mathcal{P}_r(q_n^2|\pi_r|^2)}. \quad (3.8)$$

Comparing this multiplier with $M_{q_{k+1}}$ proves (3.2).

Now suppose $k < n$ and put $r = n - k$. Again by (3.6),

$$W^{*n}W^k = W^{*r}(W^{*k}W^k) = C^{*r}M_{\overline{\pi_r}q_k} = \mathcal{P}_rM_{\overline{\pi_r}q_k}. \quad (3.9)$$

Apply Lemma 1 to (3.7) with $b = \overline{\pi_r}q_k$ and $q = q_{k+1}$. This is exactly (3.3).

For part (B), apply (3.1) to W^* . With $g = W^{*k}f$, it becomes

$$\|W^n g\|_2^2 \leq \lambda^2 \|W^* g\|_2^2. \quad (3.10)$$

Equation (3.6) gives $\|W^n g\|_2^2 = \int q_n |g|^2 d\mu$ and $W^* g = \mathcal{P}_1(\overline{\pi}g)$. Moreover,

$$\overline{\text{Ran } W^{*k}} = (\ker W^k)^\perp = L^2(S_k), \quad (3.11)$$

because $\|W^k f\|_2^2 = \int q_k |f|^2 d\mu$. Both sides of (3.10) are continuous, so (3.10) on $\text{Ran } W^{*k}$ is equivalent to (3.4) on its closure. Taking quadratic forms gives (3.5). $\square$

Corollary 3 (ordinary composition operators). *Let*

$$h_j = \frac{d(\mu \circ \varphi^{-j})}{d\mu} = \mathcal{P}_j(1).$$

Then C_φ is k -quasi n -power posinormal precisely when

$$\begin{cases} \mathcal{P}_{k-n}(h_n^2) \leq \lambda^2 h_{k+1}, & k \geq n, \\ \mathcal{P}_{n-k}\left(\frac{h_k^2}{h_{k+1}}\right) \leq \lambda^2, & k < n. \end{cases} \quad (3.12)$$

Its adjoint is k -quasi n -power posinormal precisely when

$$\int_X h_n |g|^2 d\mu \leq \lambda^2 \|\mathcal{P}_1 g\|_2^2 \quad (g \in L^2(\text{supp } h_k)). \quad (3.13)$$

Proof. Take $\pi = 1$, so that $\pi_j = 1$ and $q_j = h_j$, in Theorem 2. $\square$

When $\varphi^{-j}(\Sigma) = \Sigma$ and inverse representatives are available, $\mathcal{P}_j(a) = h_j(a \circ \varphi^{-j})$. Thus the transfer criteria collapse to the familiar pointwise expressions in Radon–Nikodym derivatives and inverse iterates. The formulas above remain valid when no inverse exists and when the conditional expectation onto $\varphi^{-j}(\Sigma)$ is nontrivial.

4 A genuinely noninvertible example

Example 4. Let $X = \{0, 1, 2\}$ with counting measure and

$$\varphi(0) = 0, \quad \varphi(1) = 0, \quad \varphi(2) = 1.$$

Then $\varphi^{-1}(\Sigma) \neq \Sigma$ and

$$h_1 = (2, 1, 0), \quad h_j = (3, 0, 0) \quad (j \geq 2).$$

For $k = 2, n = 1$, (3.12) gives

$$\mathcal{P}_1(h_1^2) = (5, 0, 0) \leq \lambda^2(3, 0, 0) = \lambda^2 h_3.$$

Consequently C_φ is 2-quasi 1-power posinormal, with optimal $\lambda^2 = 5/3$. On the other hand, for $k = n = 1$ the inequality $h_1^2 \leq \lambda^2 h_2$ fails at the point 1, so C_φ is not 1-quasi 1-power posinormal. This example lies strictly outside the source’s full-preimage sigma-algebra hypothesis.

For the adjoint and $k = 2, n = 1$, (3.13) is tested only on functions supported at 0 and its optimal constant is $\lambda^2 = 2$. For $k = 1$, the vector $g = (1, -1, 0)$ is supported on $\text{supp } h_1$, has $\mathcal{P}_1 g = 0$, and has positive h_1 -norm; hence the corresponding property fails. This also illustrates why the support compression in (3.4) is essential.

5 Verification, novelty, and limitations

The proof is analytic and does not depend on computation. As an independent check, the included verifier constructs weighted ℓ^2 matrices for 46 finite systems (43 noninjective), uses real zero/nonzero moduli and complex phases, and tests all (k, n) with $0 \leq k \leq 4$ and $1 \leq n \leq 4$. Across 920 cases it checks (3.6), both factorizations (3.8)–(3.9), the multiplier formulas, and the optimal constants from (3.2)–(3.3) against direct singular values. The recorded verdict is PASS; see `VERIFIER_REPORT.md`.

A bounded search through the run indexes and arXiv on 11 August 2026, using the exact future-work sentence, title, arXiv identifier, author combinations, and the phrases “ k -quasi n -power posinormal” and “transfer operator,” found the source, its published version, and a companion preprint, but no paper removing this hypothesis or stating Theorem 2. This is a novelty check, not a guarantee of priority.

The result addresses the first future-work direction for the composition and weighted-composition results on L^2 . It does not address the source’s other directions concerning L^p for $p \neq 2$, noncommutative spaces, Hardy/Bergman spaces, Lambert operators, directed-tree shifts, or applications. The boundedness of C_φ and the nonsingularity of φ are retained; the removed condition is exactly $\varphi^{-j}(\Sigma) = \Sigma$.

References

- [1] D. Dhara, P. Pattanayak, R. Rattan, and S. Sarkar, *k-quasi n-power posinormal Weighted Composition and Cauchy Dual of Moore–Penrose inverse of Lambert Operators*, arXiv:2507.06511 (2025), Section 6, p. 15.